SEMINAR REPORT

A seminar Report on

AI-Based Traffic Signal Optimizer

Submitted in partial fulfilment for the award of

THIRD YEAR DIPLOMA IN

COMPUTER ENGINEERING

By

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FOR THE ACADEMIC YEAR : 2026 – 2027

DEPARTMENT OF COMPUTER ENGINEERING

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code : 0059)

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(An Affiliated Institute of Maharashtra State Board of Technical Education)

DEPARTMET OF COMPUTER ENGINEERING

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TABLE OF CONTENT

Sr. No.	Chapter	Page No.
	Abstract	i
	List of Figures	ii
	List of Tables	iii
1	Introduction	1
2	Literature Review	2
3	Information/Chapters	7
4	Advantages and Disadvantages	12
5	Conclusion	13
6	Project Initiation :	14
	6.1) Description of problem statement.	14
	6.2) Scope and objectives.	15
	6.3) Stake holder	16
	6.4) Platform/ Equipment/ Resources identification.	17
7	Seminar Presentation	18
	Bibliography	28
	References	29

ABSTRACT

Traffic congestion at urban intersections causes delays, fuel wastage, increased emissions, and inconvenience for road users. Conventional traffic signals often use fixed-time schedules that may not respond effectively when vehicle volume changes across lanes. An AI-Based Traffic Signal Optimizer is a proposed intelligent system that uses traffic observations to support adaptive signal timing.

This seminar report reviews research on adaptive traffic signal control, reinforcement learning, deep reinforcement learning, multi-agent coordination, computer vision, and traffic simulation. Selected research papers are summarized and compared according to their approach, technologies, objectives, and limitations.

The proposed system can use Python, OpenCV, a vehicle-detection model such as YOLO, and a traffic simulator such as SUMO. Its performance can be evaluated using average waiting time, queue length, throughput, and travel time. Since real traffic control is safety-critical, simulation and supervised testing should precede any real-world use.

LIST OF FIGURES

Fig. No.	Title	Page No.
1	Working of AI-Based Traffic Signal Optimizer	9
2	Architecture of AI-Based Traffic Signal Optimizer	10
3	Seminar Title Page	18
4	Content Page	18
5	Abstract Page	19
6	Introduction Page	19
7	Literature Survey Page	20
8	Literature Survey Page	20
9	Literature Survey Page	21
10	Literature Survey Page	21
11	Information to Seminar Topic Page	22
12	Advantages Page	22
13	Disadvantages Page	23
14	Conclusion Page	23
15	Project Initiation	24
16	Description of Problem Statement Identification	24
17	Scope and Objective	25
18	Stack Holder Page	25
19	Seminar Exsicussion Plan	26
20	Platforms/ Equipment/ Resources	26
21	Bibliography Page	27
22	Reference Page	27

LIST OF TABLES

Table No.	Title	Page No.
1	Literature Review	5
2	Resources Required for the Project	17

CHAPTER 1

INTRODUCTION

Traffic signal control is an important part of intelligent transportation systems. At busy road intersections, vehicles arrive from different directions and compete for limited road space. If signal timing does not match current traffic demand, vehicles may wait unnecessarily, increasing travel time, fuel consumption, and emissions.

Traditional fixed-time traffic signals use predefined schedules. These schedules may not respond effectively during peak hours or when traffic demand changes suddenly. Artificial Intelligence can help analyze traffic observations and select suitable signal timings.

Computer vision can estimate vehicle counts from camera images, while reinforcement learning can learn traffic-control policies through interaction with a simulated traffic environment. The objective of an AI-Based Traffic Signal Optimizer is to improve traffic flow by adapting signal timing to observed demand while maintaining safety constraints.

The AI-Based Traffic Signal Optimizer uses artificial intelligence and real-time traffic data to adjust signal timings according to vehicle density. It helps manage traffic congestion more efficiently by allocating green-light time based on traffic demand. The system aims to reduce vehicle waiting time, fuel consumption, and harmful emissions. Simulation can be used to compare its performance with traditional fixed-time traffic signals under different traffic conditions. The proposed approach also considers road safety and appropriate signal-timing limits to ensure smooth and efficient traffic movement.

The main objectives of this project are to study traditional and AI-based traffic signal control techniques, review research papers on adaptive control and reinforcement learning, estimate traffic density or queue length for different approaches, and design an adaptive signal-timing method. The project also aims to compare the proposed method with fixed-time control using simulation and identify its limitations and possible future improvements.

CHAPTER 2
LITERATURE REVIEW

1. **IntelliLight: A Reinforcement Learning Approach for Intelligent Traffic Light Control (2018) — Wei et al.**

   The authors propose IntelliLight, a reinforcement-learning approach that selects traffic-light actions based on traffic conditions. The study investigates learning-based signal control using traffic data from surveillance cameras. It demonstrates how a controller can learn from observed traffic states rather than relying only on manually written rules. However, performance may depend on the studied traffic data and experimental setting.

2. **PressLight: Learning Max Pressure Control to Coordinate Traffic Signals in Arterial Network (2019) — Wei et al.**

   PressLight combines reinforcement learning with the max-pressure concept from traffic engineering. It represents traffic conditions using information about incoming and outgoing traffic and learns signal actions intended to improve network flow. The research is useful because it combines traffic-engineering principles with AI. However, results can depend on network layout, traffic demand, and simulation assumptions.

3. **CoLight: Learning Network-level Cooperation for Traffic Signal Control (2019) — Wei et al.**

   CoLight uses graph-attention mechanisms to help neighbouring traffic signals learn coordinated decisions. The approach considers relationships between intersections instead of treating every signal independently. It aims to improve traffic flow across larger road networks. However, computational complexity, network structure, and real-world implementation remain challenges.

4. **Deep Reinforcement Learning for Traffic Signal Control: A Review (2020) — Rasheed et al.**

   This paper reviews deep reinforcement-learning methods for traffic signal control. It discusses learning architectures, simulation platforms, traffic-control characteristics, and research challenges. The review helps explain how AI can be applied to complex traffic

conditions. Since it is a survey, it does not provide a single controller that works optimally in every traffic situation.

5. **Application of Deep Reinforcement Learning in Traffic Signal Control: An Overview and Impact of Open Traffic Data (2020)**

   This paper reviews applications of deep reinforcement learning in traffic signal control and discusses the role of open traffic data. It connects adaptive traffic management with congestion reduction and sustainable mobility. The study is useful for understanding data requirements. However, open datasets may not represent the road geometry, driving behaviour, and traffic conditions of every location.

6. **State-of-art Review of Traffic Signal Control Methods: Challenges and Opportunities (2020) — Qadri et al.**

   The authors review different traffic signal control methods and challenges caused by increasing urban traffic. The study provides a foundation for understanding conventional and advanced signal-control approaches. It highlights the need for methods that respond to changing traffic demand. However, no single method is suitable for every intersection because infrastructure and traffic conditions vary.

7. **The Traffic Signal Control Problem for Intersections: A Review (2020) — Eom and Kim**

   This paper examines intersection signal control as an optimization and scheduling problem. It discusses traffic conditions and operational factors affecting signal settings. The study shows that signal optimization requires more than simply giving additional green time to a busy road. Phase compatibility, clearance intervals, pedestrian requirements, and minimum and maximum green durations must also be considered.

8. **Recent Advances in Reinforcement Learning for Traffic Signal Control: A Survey of Models and Evaluation (2021) — Wei et al.**

   This survey categorizes reinforcement-learning approaches and reviews their strengths, limitations, simulation environments, and evaluation practices. It highlights the importance of state representation, reward functions, and experimental design. The paper is useful for selecting a methodology for an AI traffic-signal project. However, results from different papers may not be directly comparable because their networks and testing conditions differ.

9. **Reinforcement Learning in Urban Network Traffic Signal Control: A Systematic Literature Review (2022) — Noaeen et al.**

   The authors systematically review network-level reinforcement-learning research, covering 160 peer-reviewed studies within the review's defined publication period. The paper categorizes methods, simulation environments, applications, and evaluation practices. It identifies research gaps and emphasizes the importance of systematic testing. However, the findings are limited by the search period and the studies included in the review.

10. **Deep Reinforcement Learning Based Traffic Signal Control: A Comparative Analysis (2023)**

    This research compares design choices for deep reinforcement-learning traffic signal control, including state representations, reward functions, and algorithms in simulated intersection scenarios. It reports that reward-function design can strongly influence performance, especially under high traffic demand. The study also indicates that more detailed vehicle data do not automatically produce better control. Its findings need further validation for larger networks and real-world conditions.

11. **Reinforcement Learning for Adaptive Traffic Signal Control: An Overview (2026)**

    This paper reviews reinforcement-learning approaches for adaptive signal control, including single-agent and multi-agent methods. It discusses evaluation metrics and commonly used simulation platforms such as SUMO and CityFlow. The study highlights challenges related to multiple objectives, scalability, and transferring simulation results to real-world traffic. Therefore, a simulation-based student project should evaluate its approach carefully before considering practical deployment.

12. **Intelligent Traffic Signal Control Based on Reinforcement Learning: A Survey (2026)**

    This survey describes reinforcement learning as a promising approach for adapting signal timing to dynamic traffic conditions and coordinating multiple intersections. It also discusses computational costs, data requirements, and generalization across different traffic scenarios. The study supports the use of AI for adaptive traffic management, but successful real-world implementation requires local validation and safety testing.

Table No. 2.1: Literature Review

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	IntelliLight — Wei et al. (2018)	Reinforcement Learning, Deep Q-Network (DQN)	SUMO simulated traffic scenarios	Improve traffic-light decisions using real-time traffic states	Depends on training quality and simulation accuracy
2	PressLight — Wei et al. (2019)	Reinforcement Learning, Max-Pressure Control	Simulated traffic networks	Reduce congestion by considering traffic pressure	Needs reliable traffic-state information
3	CoLight — Wei et al. (2019)	Multi-Agent RL, Graph Attention Networks	Large-scale simulated road networks	Coordinate traffic signals across multiple intersections	Coordination becomes complex in large networks
4	FRAP — Zheng et al. (2019), Learning Phase Competition for Traffic Signal Control	Phase Competition Model, Reinforcement Learning	Traffic simulation scenarios	Select suitable signal phases based on competing traffic movements	Performance may vary under unfamiliar traffic patterns
5	Chu et al. (2019), Multi-Agent Deep Reinforcement Learning for Large-Scale Traffic Signal Control	Multi-Agent RL, Advantage Actor-Critic (A2C)	Synthetic traffic grid and Monaco network simulations	Improve scalable control of multiple intersections	Limited local observations and coordination challenges
6	Cooperative Deep Q-Learning With Q-Value Transfer for Multi-Intersection Signal Control (2019)	Deep Q-Learning, Multi-Agent RL	Simulated multi-intersection traffic	Improve cooperation and knowledge sharing between intersections	Training and knowledge transfer may be difficult across different networks

7	Rasheed et al. (2020), Deep Reinforcement Learning for Traffic Signal Control: A Review	Deep Reinforcement Learning, neural networks	Published research studies	Review DRL methods, simulators, and challenges	Findings depend on the studies and evaluation methods reviewed
8	Miletic et al. (2020), Application of Deep Reinforcement Learning in Traffic Signal Control	Deep RL, Deep Neural Networks, traffic-data analysis	Open traffic data and published studies	Examine DRL methods and the importance of open traffic data	Data availability and quality can limit real-world adoption
9	Qadri et al. (2020), State-of-Art Review of Traffic Signal Control Methods	Optimization, computational intelligence, microsimulation	Published studies and reviewed simulation methods	Compare signal-timing optimization approaches	Different methods may perform differently across traffic conditions
10	Eom and Kim (2020), The Traffic Signal Control Problem for Intersections: A Review	Traffic-control algorithms, optimization, simulation	Previously published intersection-control studies	Review signal-control problems, methods, and evaluation practices	Practical constraints and complex traffic conditions remain challenging
11	Wang et al. (2021), Adaptive Traffic Signal Control for Large-Scale Scenario with Cooperative Group-Based Multi-Agent Reinforcement Learning	Multi-Agent RL, cooperative learning, SUMO	Synthetic network, Monaco and Harbin traffic simulations	Coordinate signals and improve congestion management in large networks	Requires effective coordination and substantial training
12	Noaeen et al. (2022), Reinforcement Learning in Urban Network Traffic Signal Control: A Systematic Literature Review	Reinforcement Learning, multi-agent traffic control	160 reviewed research articles	Analyze RL methods and research gaps in urban traffic networks	Different evaluation setups make comparisons difficult

CHAPTER 3
INFORMATION/CHAPTERS

1. Overview of AI-Based Traffic Signal Optimizer

An AI-Based Traffic Signal Optimizer is an intelligent traffic-management approach that adjusts traffic signal timing according to observed or estimated traffic conditions. Unlike a conventional fixed-time signal, the system can use traffic information to decide which direction should receive a green signal and for how long, within predefined limits.

The system may use cameras, vehicle detectors, or simulated traffic information as input. Computer vision can estimate the number of vehicles in each lane, while a control algorithm uses these measurements to choose the next signal phase. The aim is to reduce unnecessary waiting, manage queue lengths, and improve the movement of vehicles through an intersection.

The exact performance of the system depends on its algorithm, quality of traffic observations, intersection layout, and operating conditions. Therefore, the system should be evaluated using simulation or controlled testing before practical deployment.

2. Types of Traffic Signal Control

1. Fixed-Time Traffic Signal Control

Fixed-time control uses predefined green, yellow, and red durations. The schedule may repeat at regular intervals. It is comparatively simple but may not respond well when the amount of traffic changes.

2. Vehicle-Actuated Traffic Signal Control

Vehicle-actuated control uses detectors to identify vehicles or demand at an intersection. The signal may extend or terminate a phase according to detector input. Its flexibility depends on the detector arrangement and control logic.

3. Adaptive Traffic Signal Control

Adaptive control uses current traffic observations to adjust signal timing. It attempts to match signal operation with changing demand rather than depending entirely on a fixed schedule.

4. AI-Based Traffic Signal Control

AI-based control applies techniques such as machine learning, computer vision, or reinforcement learning to analyse traffic conditions and select signal actions. Some systems optimise a single intersection, while others coordinate multiple intersections.

3. Need for AI-Based Traffic Signal Optimisation

The need for intelligent traffic signal optimisation arises from several traffic-management challenges:

1. Increasing Traffic Volume: Growing numbers of vehicles can create longer queues at busy intersections.
2. Unequal Traffic Demand: Different roads may experience different traffic volumes at the same time.
3. Fixed Signal Timings: Predefined timings may not suit every traffic situation.
4. Unnecessary Waiting: Vehicles may wait even when the opposing road has relatively low demand.
5. Fuel Consumption and Emissions: Repeated stopping and idling can contribute to fuel use and emissions.
6. Changing Traffic Conditions: Accidents, roadworks, peak hours, and special events can alter traffic flow.
7. Network Coordination: Congestion at one intersection may affect neighbouring roads.

4. Working Principle

The working process consists of the following steps:

1. Start: The system starts the traffic monitoring or simulation process.
2. Traffic Data Collection: It receives traffic information from cameras, sensors, or a simulator.
3. Data Pre-processing: The collected information is prepared for further analysis.
4. Vehicle Detection: A computer-vision model or detector estimates vehicle counts and queue lengths.
5. Traffic-State Estimation: The system identifies the current demand for each approach to the intersection.
6. AI-Based Optimisation: The algorithm selects an appropriate signal phase or timing action.

7. Safety Check: The proposed action is checked against signal-sequence rules and minimum or maximum timing constraints.
8. Signal Timing Update: The selected action is applied within the authorised control environment.
9. Performance Monitoring: The system measures waiting time, queue length, throughput, or average delay.
10. Feedback: Updated traffic conditions are used for the next decision.
11. End: The process ends when monitoring stops; otherwise, the cycle continues.

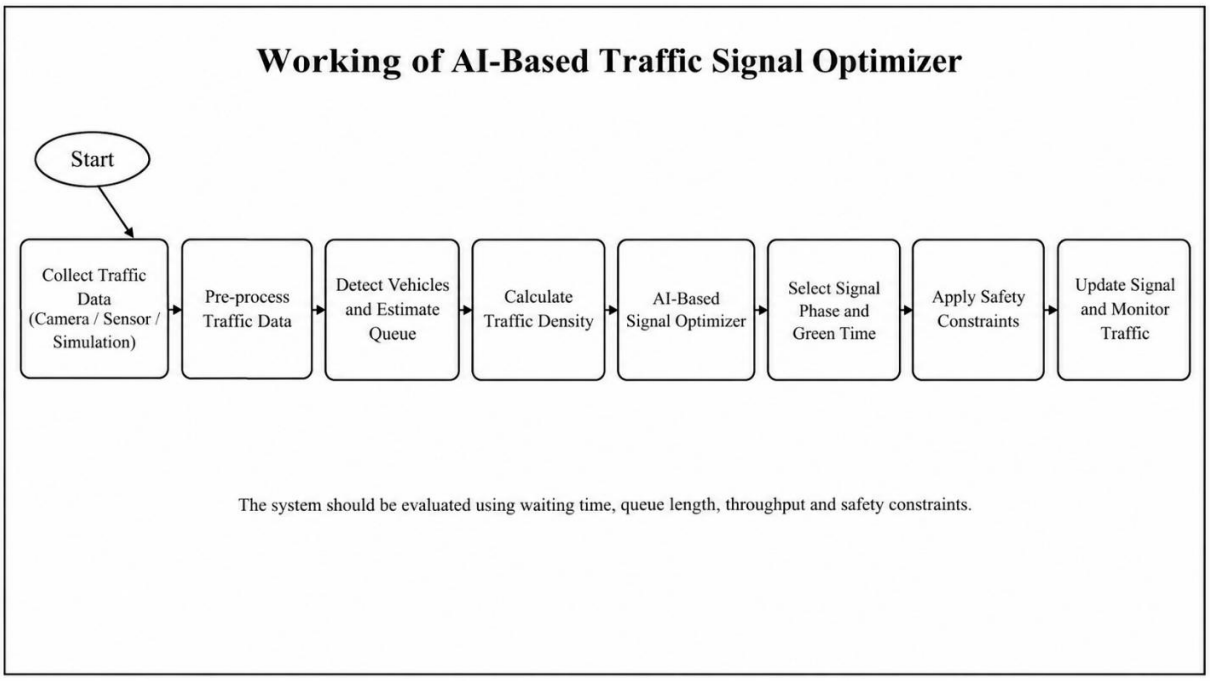


Fig. 3.1: Working of AI-Based Traffic Signal Optimizer

5. System Architecture and Main Components

The architecture consists of the following main components:

1. Traffic Environment: Represents the actual road intersection or simulated road network.
2. Data Collection Module: Collects vehicle information from cameras, sensors, or a traffic simulator.
3. Pre-processing Module: Prepares the observations for analysis and reduces unsuitable input data.
4. Traffic-State Estimation Module: Estimates vehicle count, queue length, waiting time, and traffic demand.

5. AI Optimisation Engine: Selects signal-control actions based on traffic conditions and the selected algorithm.
6. Safety and Timing Module: Ensures that selected actions follow approved signal sequences and timing limits.
7. Signal Control Module: Sends or applies signal actions within the test or authorised control environment.
8. Monitoring Dashboard: Displays traffic observations, signal status, and evaluation results.

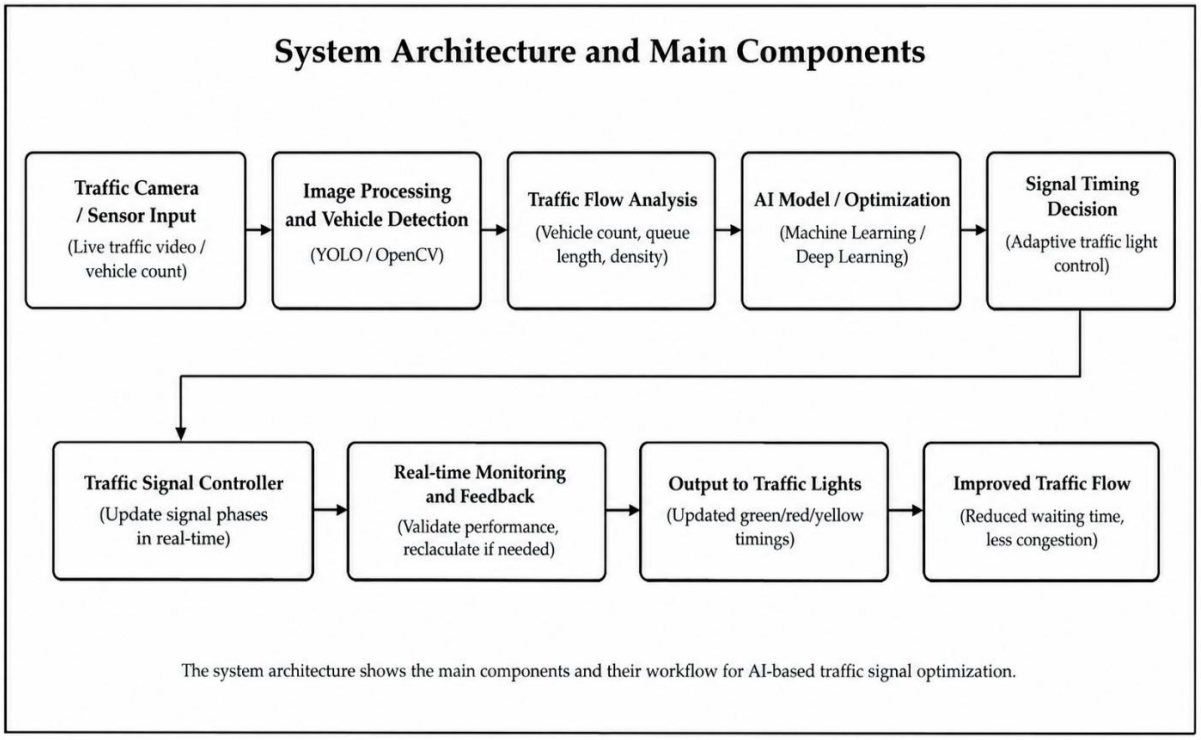


Fig. 3.2: Architecture of AI-Based Traffic Signal Optimizer

6. Technologies and Techniques Used

Python is used to implement AI algorithms and traffic-processing logic. OpenCV helps process traffic images and videos, while YOLO detects and counts vehicles. Machine Learning predicts traffic demand, and Reinforcement Learning helps optimize traffic signal timings. SUMO simulates road networks, vehicles, and traffic signals, while TraCI connects external programs to the simulation. Matplotlib or web dashboards visualize traffic data and performance. A database can store traffic observations and signal decisions. The technology selection depends on project requirements and available hardware.

7. Example of AI-Based Traffic Signal Optimisation

Consider a four-way road intersection with traffic coming from the north, south, east, and west. During morning peak hours, the north–south road may have more vehicles than the east–west road. A fixed-time signal provides the same green-light duration to both directions, whereas an AI-based optimiser analyses traffic conditions and adjusts the green-light duration according to vehicle demand and safety constraints.

When the queue decreases, the system analyses traffic from other directions and adjusts the next signal phase accordingly. This process continues as new traffic data arrives. The system's performance can be evaluated in a simulation by comparing average waiting time, queue length, throughput, and travel time with a fixed-time signal system.

8. Performance Evaluation

The proposed system should be evaluated using clearly defined performance measures. Average Waiting Time measures the time vehicles spend waiting at an intersection, while Average Queue Length measures the number of vehicles waiting. Average Travel Time calculates the time required to travel through the selected network, and Throughput measures the number of vehicles passing through during a defined period. Average Delay measures additional travel time compared with a reference condition. Emissions Estimate evaluates fuel consumption or emissions when supported by the simulator, while Safety Compliance checks whether the system follows proper signal sequencing, yellow-light intervals, and other safety constraints.

CHAPTER 4
ADVANTAGES AND DISADVANTAGES

4.1 Advantages:

1. Adaptive Signal Timing: The system can adjust signal timing based on changing traffic demand instead of relying only on fixed schedules.
2. Potential Reduction in Waiting Time: By responding to queue lengths and traffic conditions, the system may reduce unnecessary waiting when the algorithm is effective.
3. Improved Traffic Flow: Better timing decisions may help vehicles move through intersections more efficiently.
4. Traffic Monitoring: Camera-based or detector-based systems can provide information about vehicle counts and queue conditions.
5. Support for Multiple Intersections: Advanced AI methods can coordinate neighbouring traffic signals to manage traffic across a network.
6. Performance Analysis: Simulation tools allow the system to be compared with existing control strategies using measurable traffic indicators.

4.2 Disadvantages:

1. High Initial Cost: Cameras, detectors, computing hardware, communication equipment, and installation may increase implementation cost.
2. Dependence on Data Quality: Poor camera views, occlusion, weather, or detector failures can produce inaccurate traffic estimates.
3. Computational Requirements: Some AI algorithms require considerable computing power and training time.
4. Complex Implementation: Integrating AI software with existing traffic controllers and road infrastructure can be difficult.
5. Safety Concerns: Incorrect signal decisions may disrupt traffic or create unsafe situations if proper constraints and testing are absent.
6. Limited Generalisation: An algorithm trained or evaluated on one intersection may not perform equally well on a different road layout or traffic pattern.

CHAPTER 5

CONCLUSION

AI-Based Traffic Signal Optimisation is a promising application of artificial intelligence in intelligent transportation systems. Traditional fixed-time traffic signals use predefined schedules that may not respond efficiently to changing traffic conditions. AI-based methods analyse vehicle counts, queue lengths, waiting times, and other traffic information to select suitable signal timings. Research includes adaptive scheduling, deep reinforcement learning, cooperative traffic signal control, and reinforcement-learning techniques. However, their performance depends on traffic data, road-network structure, algorithm design, and operational constraints.

The proposed system can collect traffic information, estimate traffic conditions, select suitable signal actions, check safety constraints, and monitor performance. A simulator such as SUMO can be used to test the system and compare it with fixed-time signal control. Major challenges include accurate vehicle detection, computational requirements, coordination between intersections, and applying simulation results to real roads. Therefore, careful testing and established traffic-engineering practices are essential. With proper evaluation and responsible implementation, AI-based traffic signal optimisation may help improve urban traffic management.

CHAPTER 6
PROJECT INITIATION

6.1 Description of Problem Statement

Traffic congestion at road intersections is a common transportation problem. Traditional fixed-time traffic signals operate according to predefined schedules and may not respond effectively when vehicle demand changes between different roads. As a result, vehicles may experience unnecessary waiting, longer queues, increased travel time, and additional fuel consumption.

Existing adaptive systems can use traffic observations to adjust signal timings, but practical implementation may be affected by sensor accuracy, changing traffic patterns, computing requirements, safety constraints, and infrastructure costs. AI-based methods provide an opportunity to analyse traffic conditions and select signal-control actions more flexibly.

The proposed project aims to study and design an AI-Based Traffic Signal Optimizer that estimates traffic demand and selects suitable signal timings within predefined safety limits. A prototype may be evaluated in a traffic simulator using measures such as average waiting time, queue length, throughput, and average travel time. The project will compare its performance with a fixed-time baseline under defined test conditions. The proposed system will not be considered ready for public-road deployment without further testing and approval.

6.2 Scope of the Project

The scope of the project is to study and develop a prototype of an AI-based traffic signal optimisation system for a selected intersection or small simulated road network.

The project may include traffic-data collection from a simulator or sample video, vehicle-count estimation, traffic-state processing, signal-timing decisions, and performance measurement. A rule-based controller can be implemented as a baseline, followed by an AI-based approach where resources and time permit.

The system can be tested under different traffic-demand conditions, including low, medium, and high traffic volumes. The results can be compared with fixed-time signal control using selected performance metrics.

The initial scope is limited to academic research, simulation, or controlled demonstration. Direct control of public-road traffic signals, guaranteed accident prevention, and guaranteed congestion reduction are outside the scope of the initial prototype.

Objectives of the Project

1. To study the working of conventional and adaptive traffic signal systems.
2. To review research on AI-based traffic signal optimisation.
3. To understand how vehicle count, queue length, and waiting time can be used to estimate traffic demand.
4. To design a basic architecture for an AI-based signal optimiser.
5. To explore suitable AI methods, including machine learning and reinforcement learning.
6. To evaluate the proposed approach in a simulation or controlled test environment.
7. To compare the proposed approach with fixed-time control using defined traffic metrics.
8. To identify implementation challenges involving data quality, safety, computing resources, and real-world deployment.

6.3 Stakeholders

The stakeholders are the individuals and groups who develop, evaluate, operate, or may be affected by an AI-based traffic signal optimisation system.

1. Road Users: Drivers, passengers, cyclists, and pedestrians who use roads and intersections and may be affected by signal timing.
2. Traffic Police and Traffic Authorities: Authorities responsible for traffic management, signal operation, road safety, and approval of any real-world implementation.
3. Project Developers: The team responsible for designing the software, processing traffic data, implementing the optimisation algorithm, and testing the prototype.
4. Transport Planners and Engineers: Professionals who study road layouts, traffic patterns, signal timing, and transportation efficiency.

5. Municipal Authorities: Local government bodies that may manage traffic infrastructure and consider intelligent transportation projects.
6. Researchers and Students: Individuals who study AI algorithms, computer vision, traffic simulation, and evaluation methods.
7. Project Guide and Evaluators: Faculty members who guide the seminar, review the report, and evaluate the proposed project design.

6.4 Platform / Equipment / Resources Identification

1. Platform

The project can be developed using Windows or Linux. Visual Studio Code can be used for coding and debugging. Python can implement traffic processing and AI algorithms. SUMO can provide a simulated traffic environment, while OpenCV can support image or video processing if camera-based traffic estimation is included.

For an initial prototype, simulation is a practical approach because it allows different traffic scenarios to be tested without directly controlling public-road signals. A GPU may be helpful if a large vehicle-detection model or computationally demanding AI method is selected, but it is not mandatory for every prototype.

2. Equipment

- Computer or Laptop: Required for coding, simulation, and testing.
- Processor: A modern multi-core processor suitable for general development and traffic simulation.
- RAM: At least 8 GB is a practical starting point for a basic prototype; larger simulations may require more.
- GPU (Optional): Can accelerate deep-learning training or inference when a suitable model is used.
- Internet Connection: Needed to obtain permitted software packages, documentation, and research papers.
- Storage: Required for software, simulation scenarios, datasets, videos, and experimental results.
- Camera (Optional): Useful if the project includes live or recorded camera-based vehicle detection rather than simulation-only input.

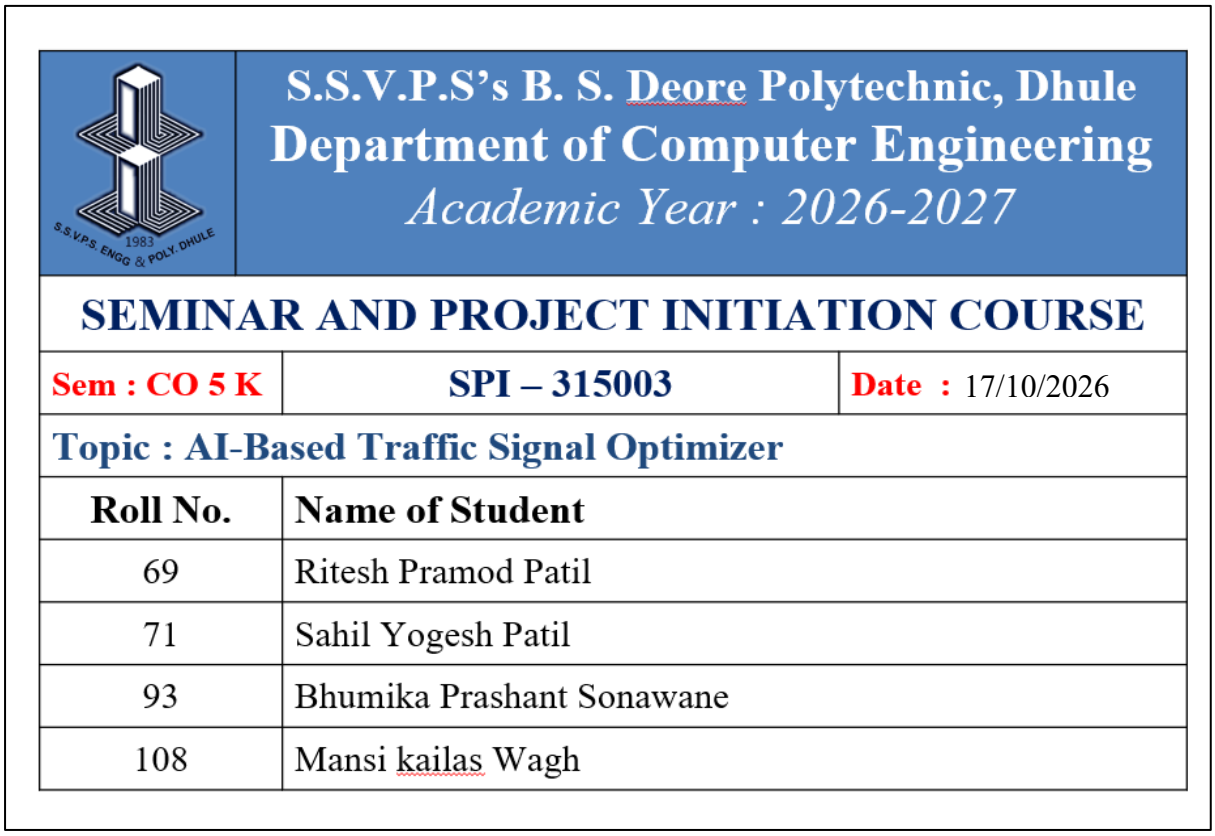
3. Resources

Table No. 6.1: Resources

Resources	Name	Purpose
Operating System	Windows 11 / Linux	Hosts development tools and the testing environment
Programming Language	Python 3.x	Implements data processing, AI algorithms, and evaluation scripts
Code Editor	Visual Studio Code	Coding, debugging, and project organisation
Traffic Simulator	SUMO	Models vehicles, roads, intersections, and traffic signals
Simulation Interface	TraCI	Allows Python code to interact with a SUMO simulation
Image Processing	OpenCV	Processes traffic images and video frames
Vehicle Detection	YOLO or another suitable model (optional)	Estimates vehicle counts from camera images
AI Framework	PyTorch or TensorFlow (optional)	Supports deep-learning models when required
Data Storage	CSV files or a database	Stores experiment settings and performance results
Hardware	Laptop; GPU if available	Runs simulation, development, and optional model training
Research Resources	Research papers and documentation	Supports literature review and system design

CHAPTER 7

SEMINAR PRESENTATION



S.S.V.P.S's B. S. Deore Polytechnic, Dhule
Department of Computer Engineering
Academic Year : 2026-2027

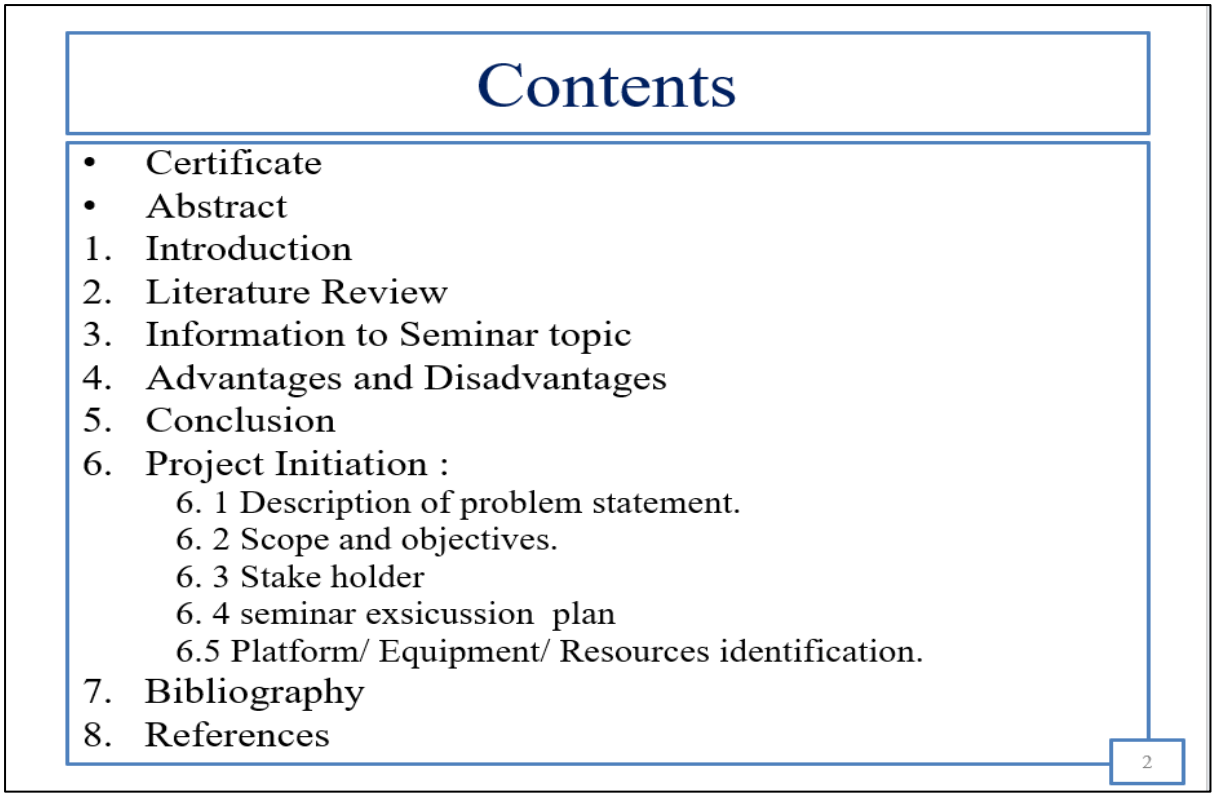
SEMINAR AND PROJECT INITIATION COURSE

Sem : CO 5 K	SPI – 315003	Date : 17/10/2026

Topic : AI-Based Traffic Signal Optimizer

Roll No.	Name of Student
69	Ritesh Pramod Patil
71	Sahil Yogesh Patil
93	Bhumika Prashant Sonawane
108	Mansi kailas Wagh

Fig No. 3: Seminar Title Page



Contents

- Certificate
- Abstract

1. Introduction
2. Literature Review
3. Information to Seminar topic
4. Advantages and Disadvantages
5. Conclusion
6. Project Initiation :
   - 6. 1 Description of problem statement.
   - 6. 2 Scope and objectives.
   - 6. 3 Stake holder
   - 6. 4 seminar exsicussion plan
   - 6.5 Platform/ Equipment/ Resources identification.
7. Bibliography
8. References

2

Fig No. 4: Content Page

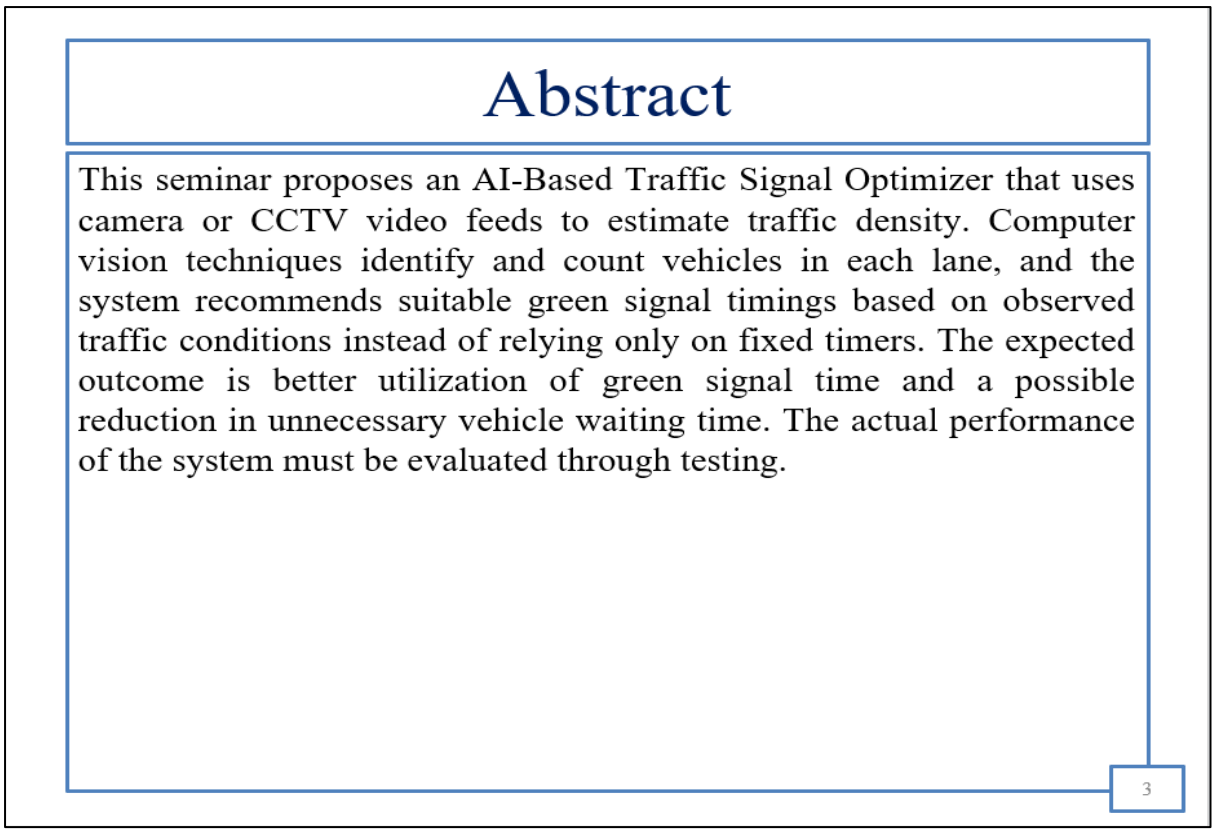


Fig No. 5: Abstract Page

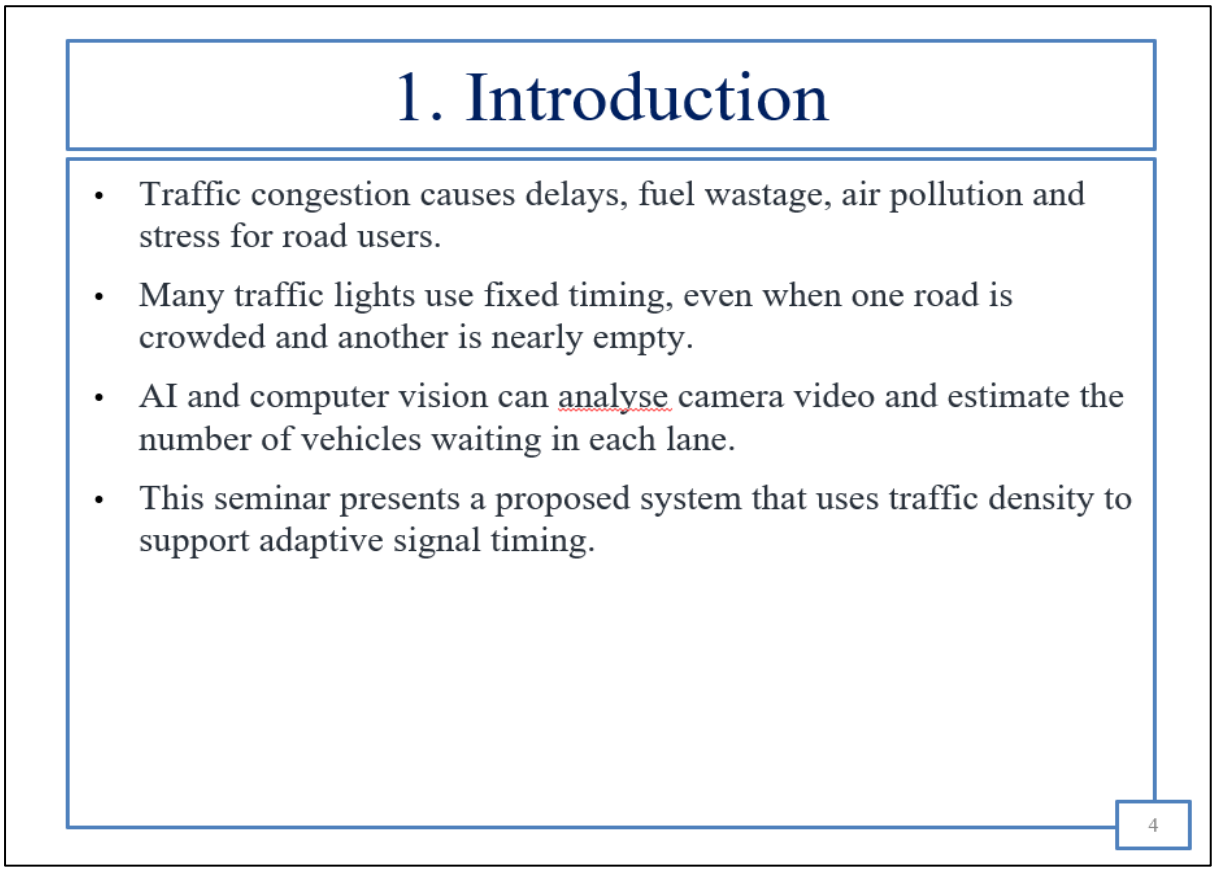


Fig No. 6: Introduction Page

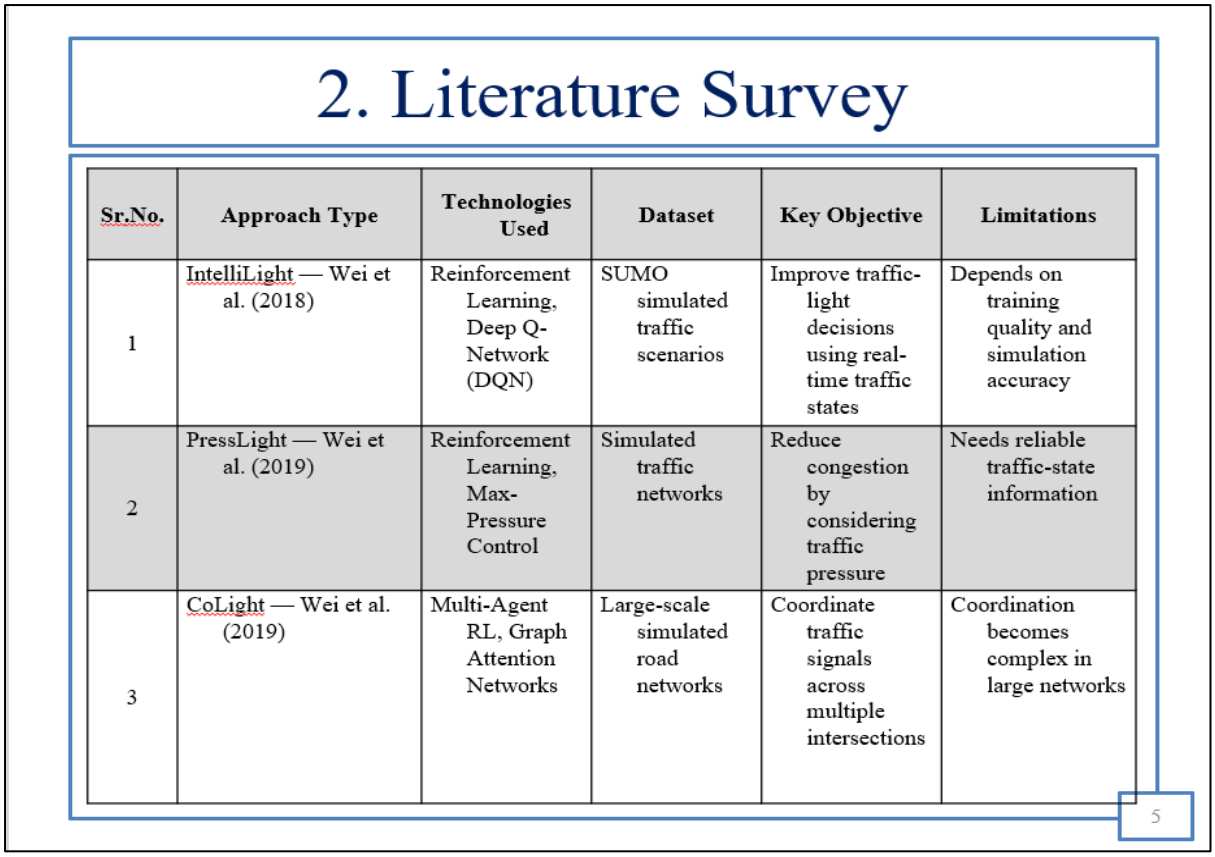


Fig No. 7: Literature Survey

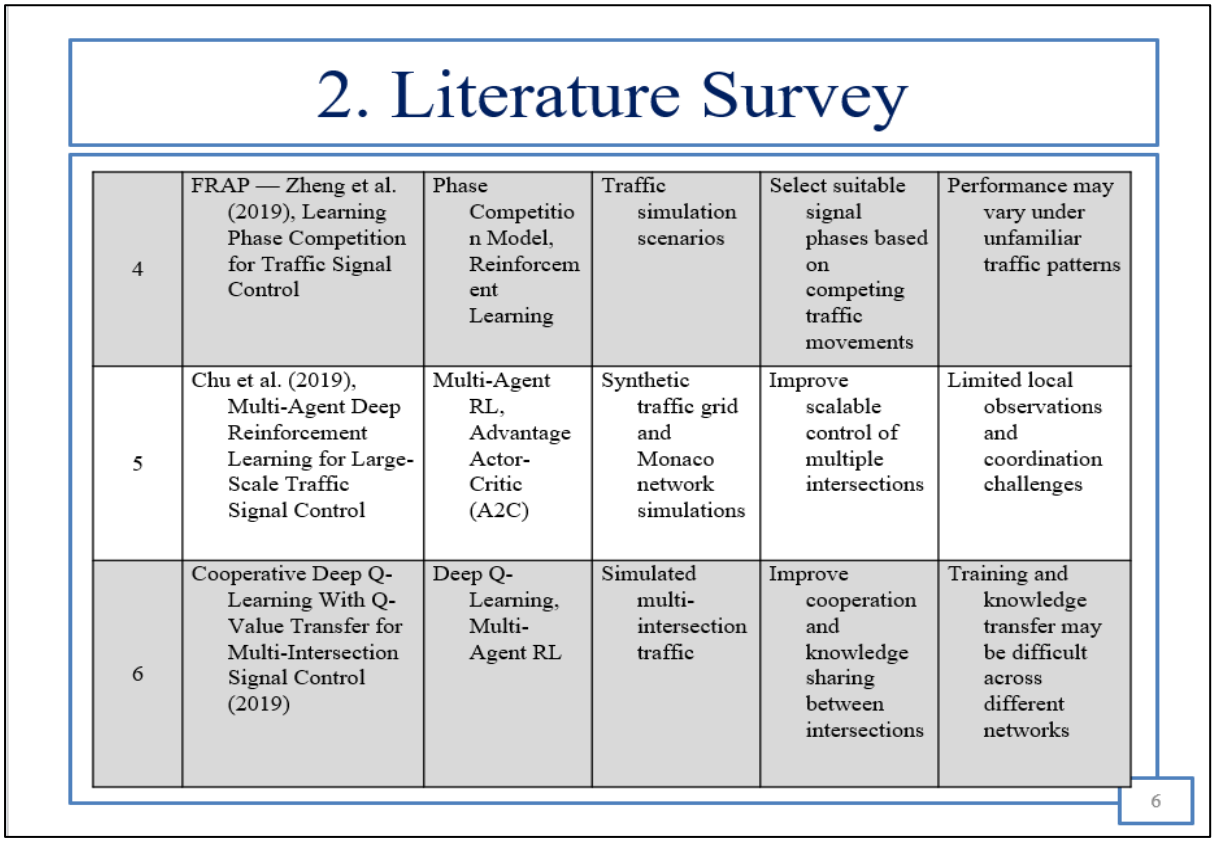
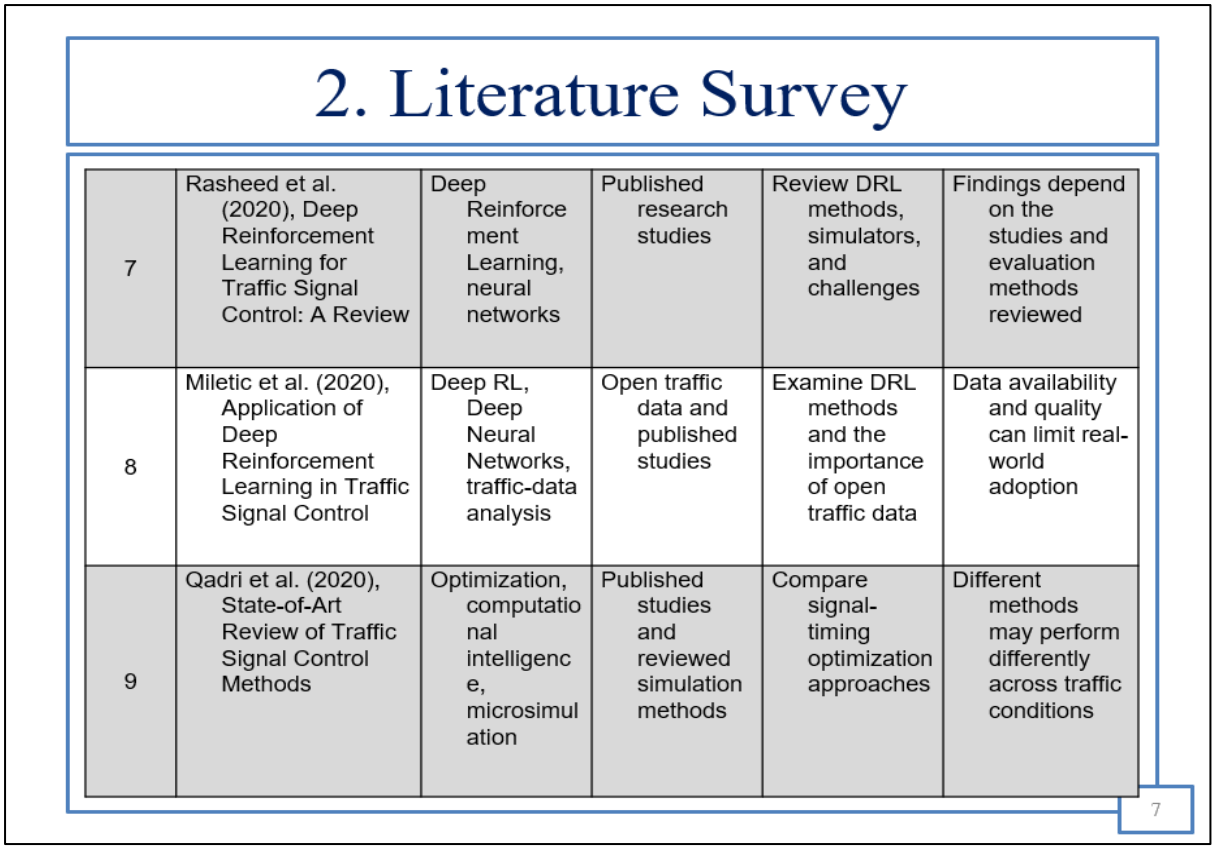


Fig No. 8: Literature Survey

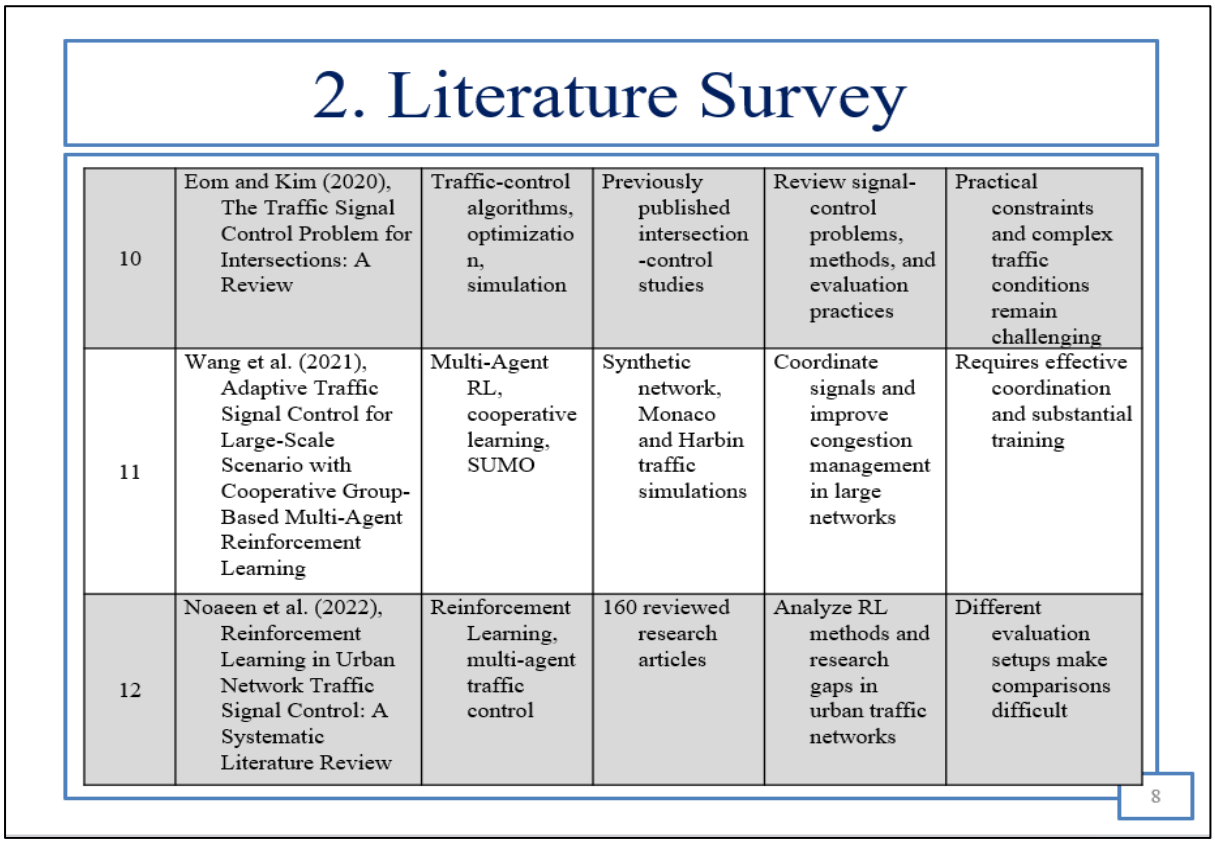


2. Literature Survey

7	Rasheed et al. (2020), Deep Reinforcement Learning for Traffic Signal Control: A Review	Deep Reinforcement Learning, neural networks	Published research studies	Review DRL methods, simulators, and challenges	Findings depend on the studies and evaluation methods reviewed
8	Miletic et al. (2020), Application of Deep Reinforcement Learning in Traffic Signal Control	Deep RL, Deep Neural Networks, traffic-data analysis	Open traffic data and published studies	Examine DRL methods and the importance of open traffic data	Data availability and quality can limit real-world adoption
9	Qadri et al. (2020), State-of-Art Review of Traffic Signal Control Methods	Optimization, computational intelligence, microsimulation	Published studies and reviewed simulation methods	Compare signal-timing optimization approaches	Different methods may perform differently across traffic conditions

7

Fig No. 9: Literature Survey



2. Literature Survey

10	Eom and Kim (2020), The Traffic Signal Control Problem for Intersections: A Review	Traffic-control algorithms, optimization, simulation	Previously published intersection-control studies	Review signal-control problems, methods, and evaluation practices	Practical constraints and complex traffic conditions remain challenging
11	Wang et al. (2021), Adaptive Traffic Signal Control for Large-Scale Scenario with Cooperative Group-Based Multi-Agent Reinforcement Learning	Multi-Agent RL, cooperative learning, SUMO	Synthetic network, Monaco and Harbin traffic simulations	Coordinate signals and improve congestion management in large networks	Requires effective coordination and substantial training
12	Noaeen et al. (2022), Reinforcement Learning in Urban Network Traffic Signal Control: A Systematic Literature Review	Reinforcement Learning, multi-agent traffic control	160 reviewed research articles	Analyze RL methods and research gaps in urban traffic networks	Different evaluation setups make comparisons difficult

8

Fig No. 10: Literature Survey

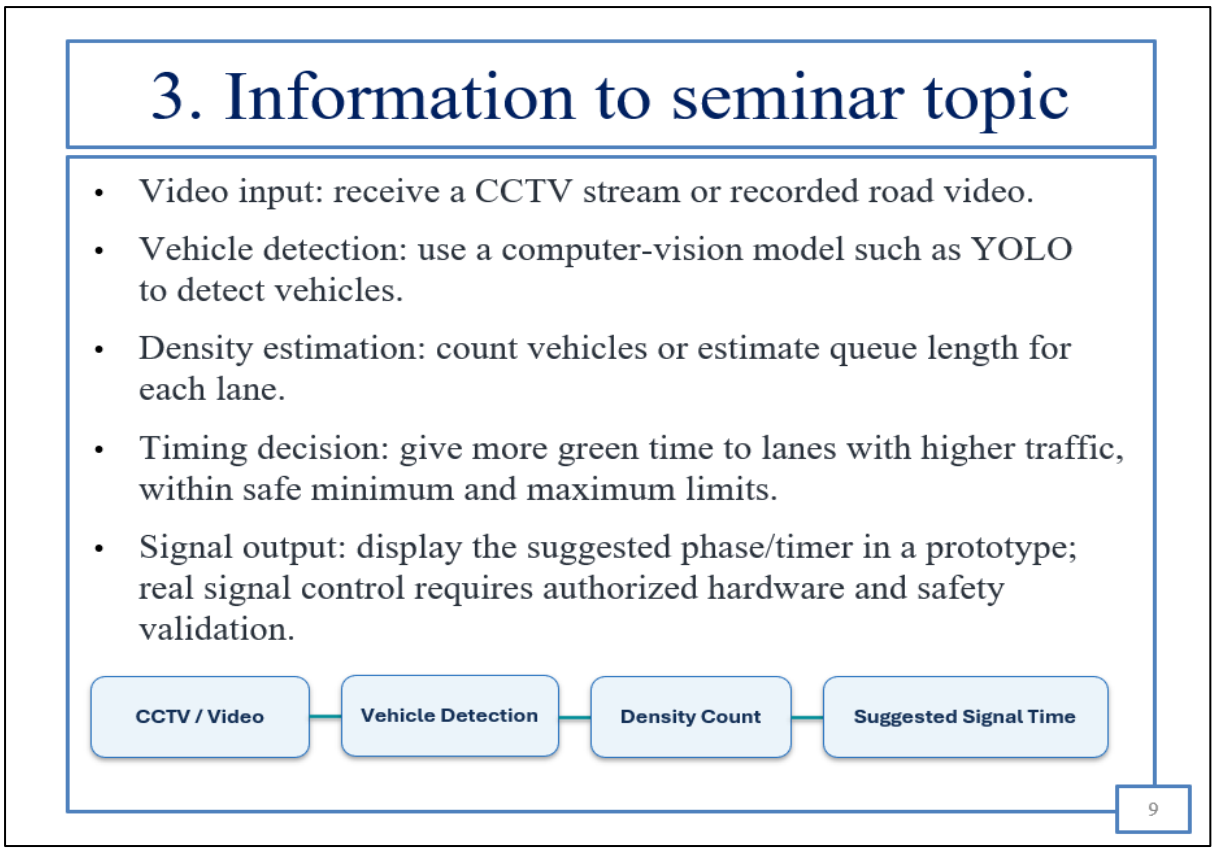


Fig No. 11: Information to Seminar Topic

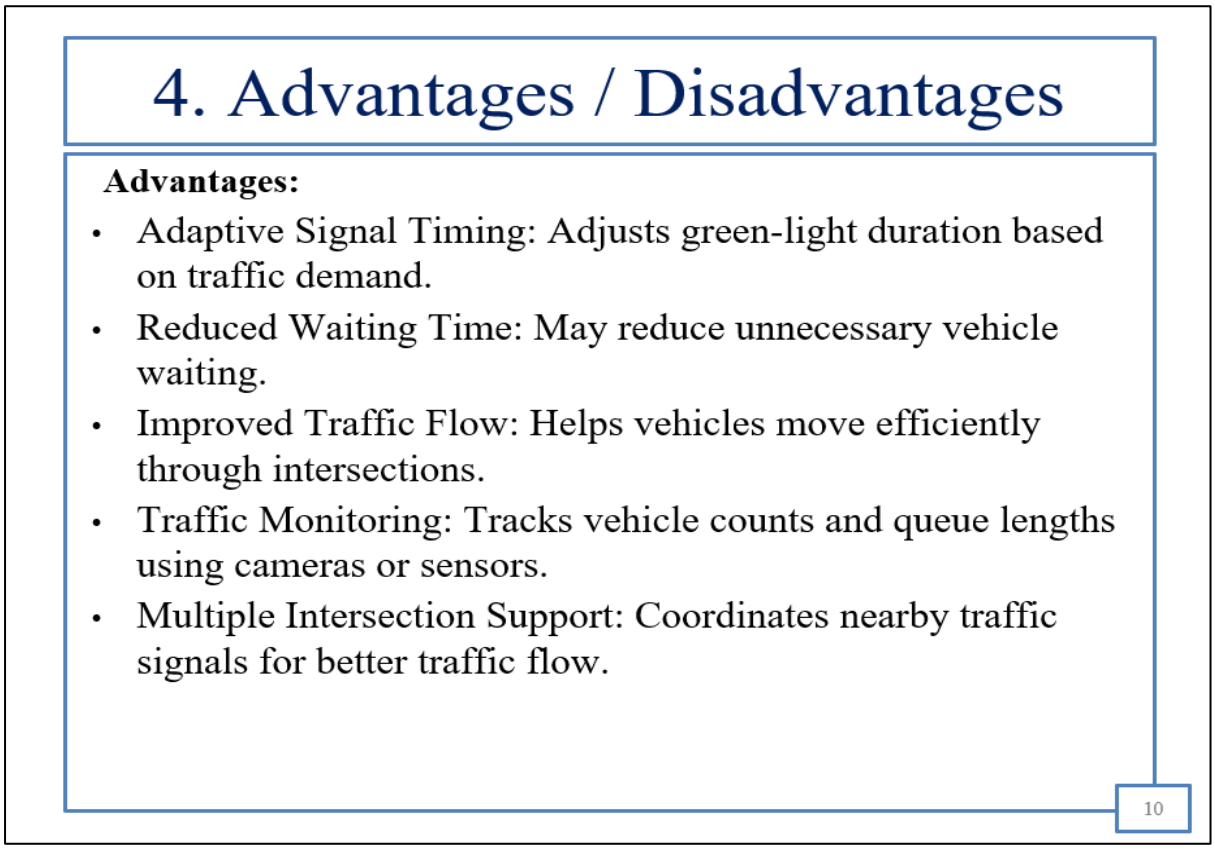


Fig No. 12: Advantages Page

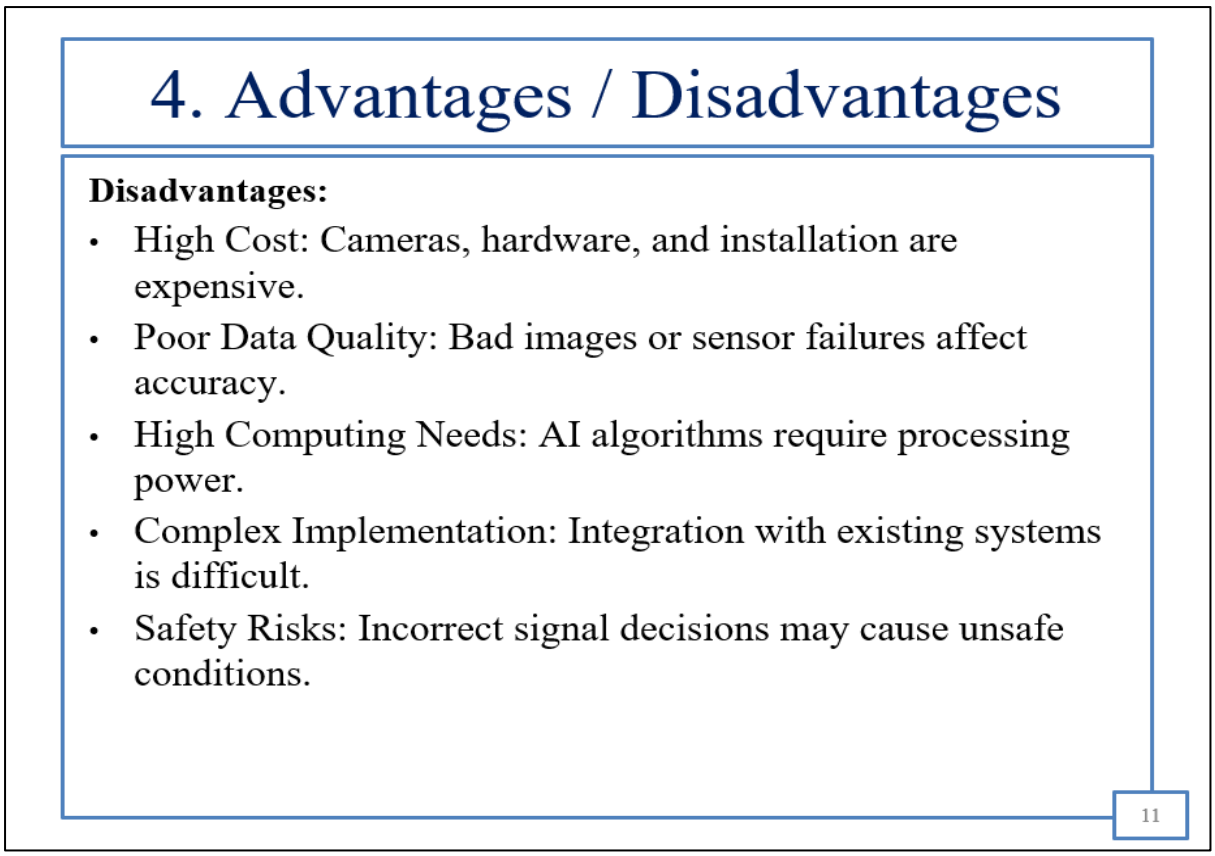


Fig No. 13: Disadvantages Page

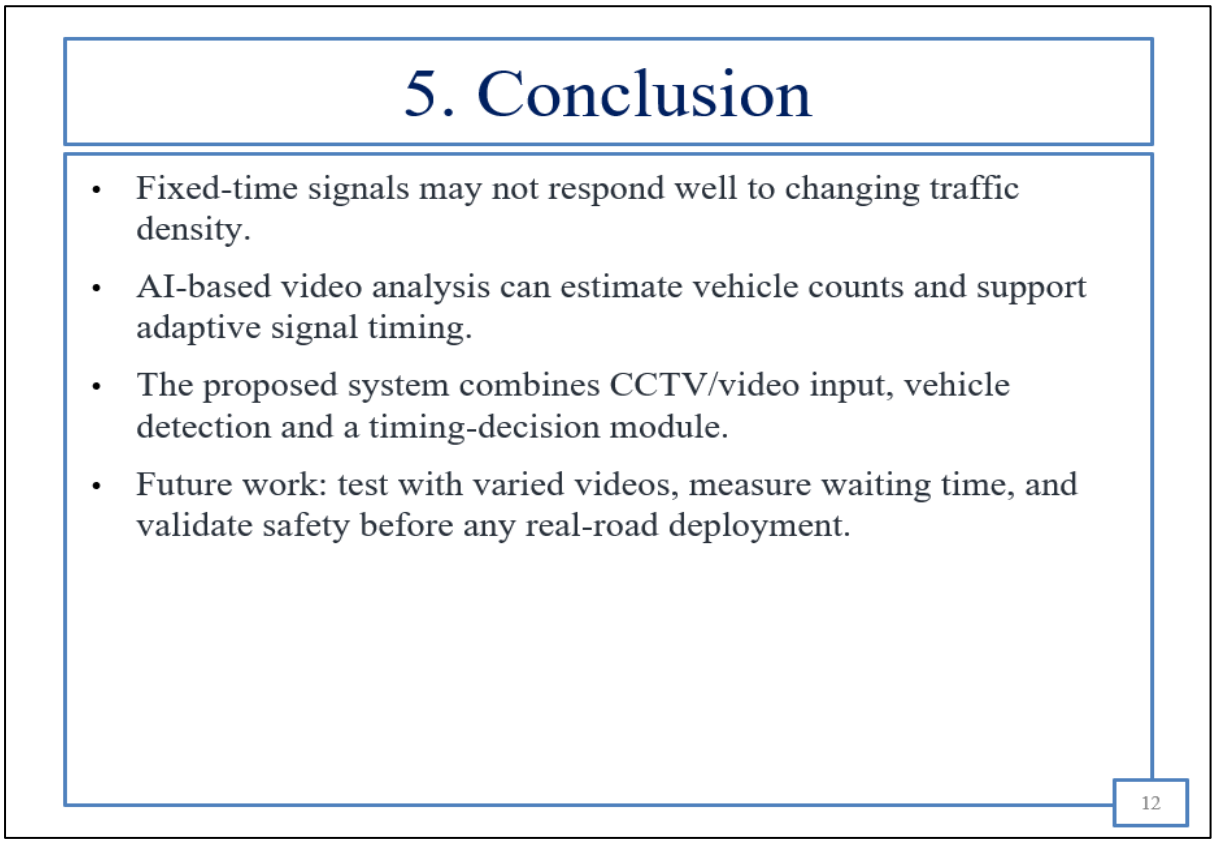


Fig No. 14: Conclusion Page

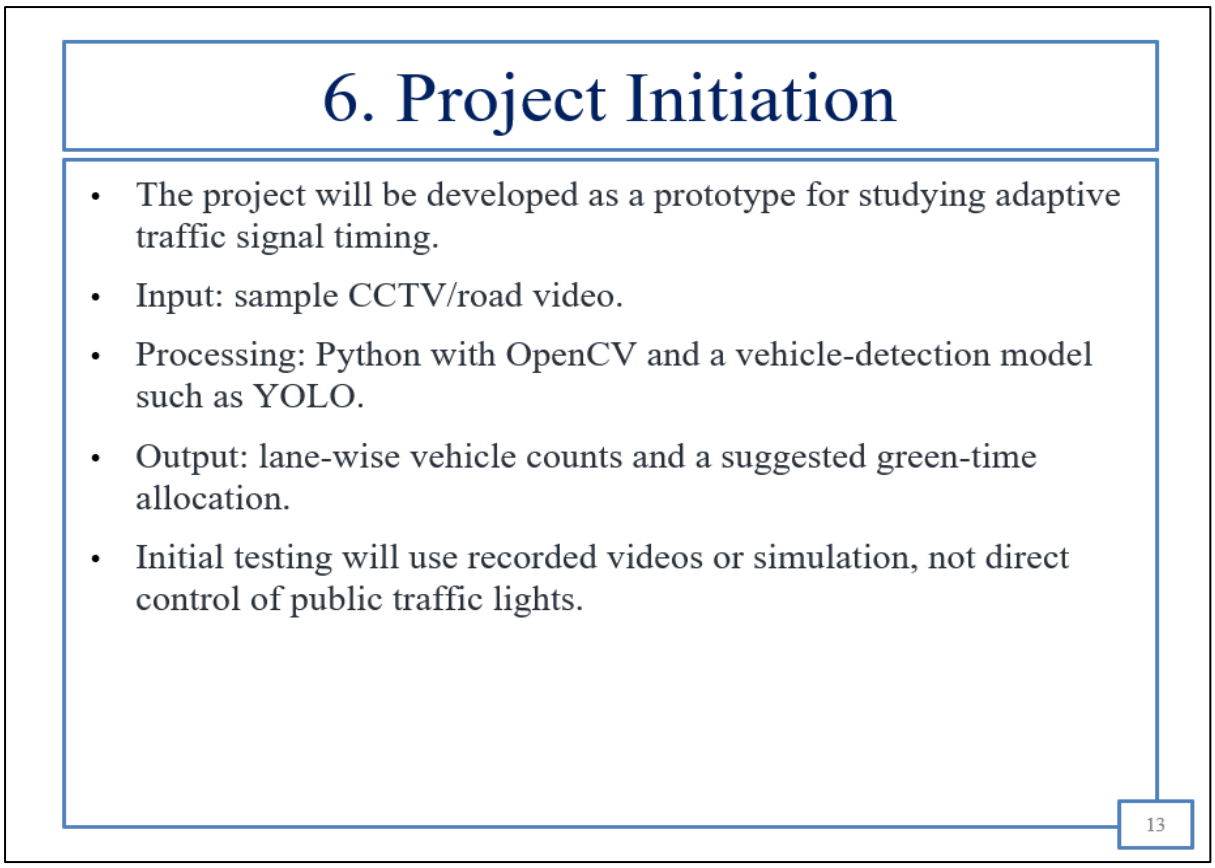


Fig No. 15: Project Initiation

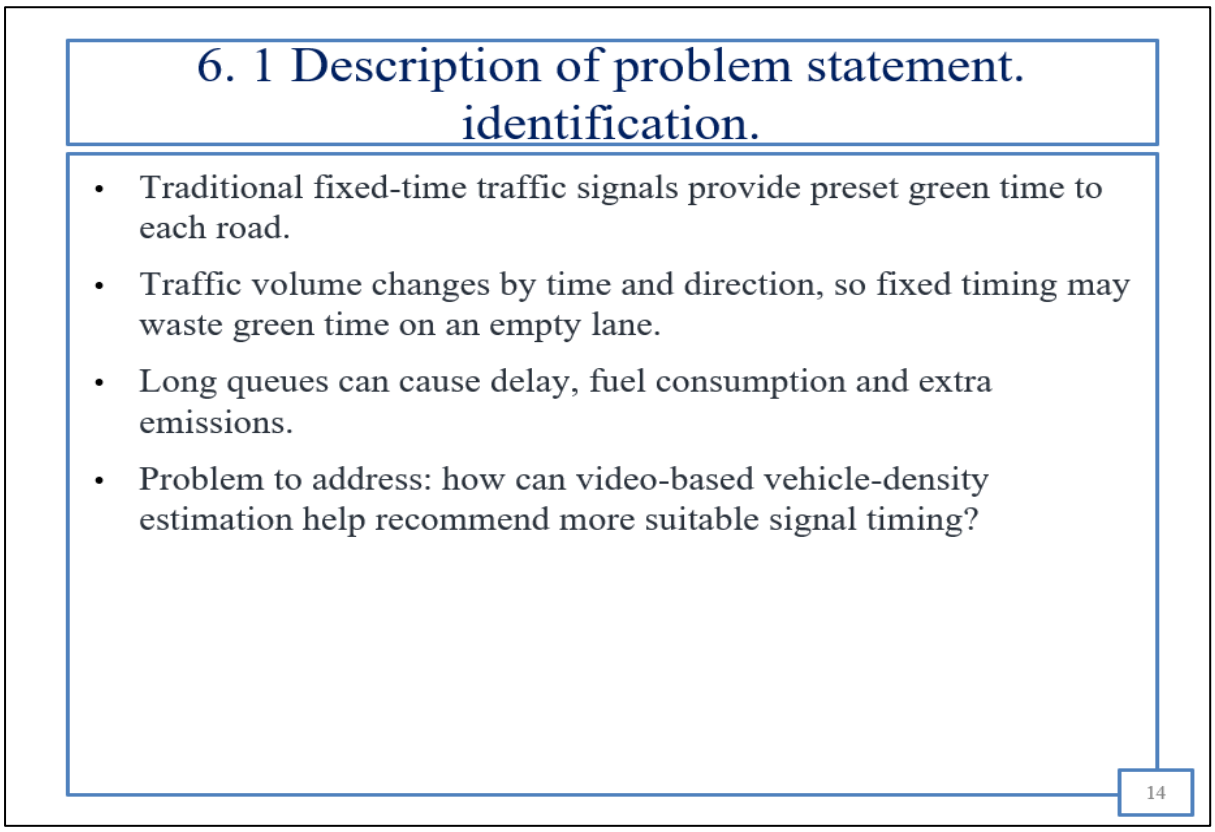


Fig No. 16: Description of Problem Statement Identification

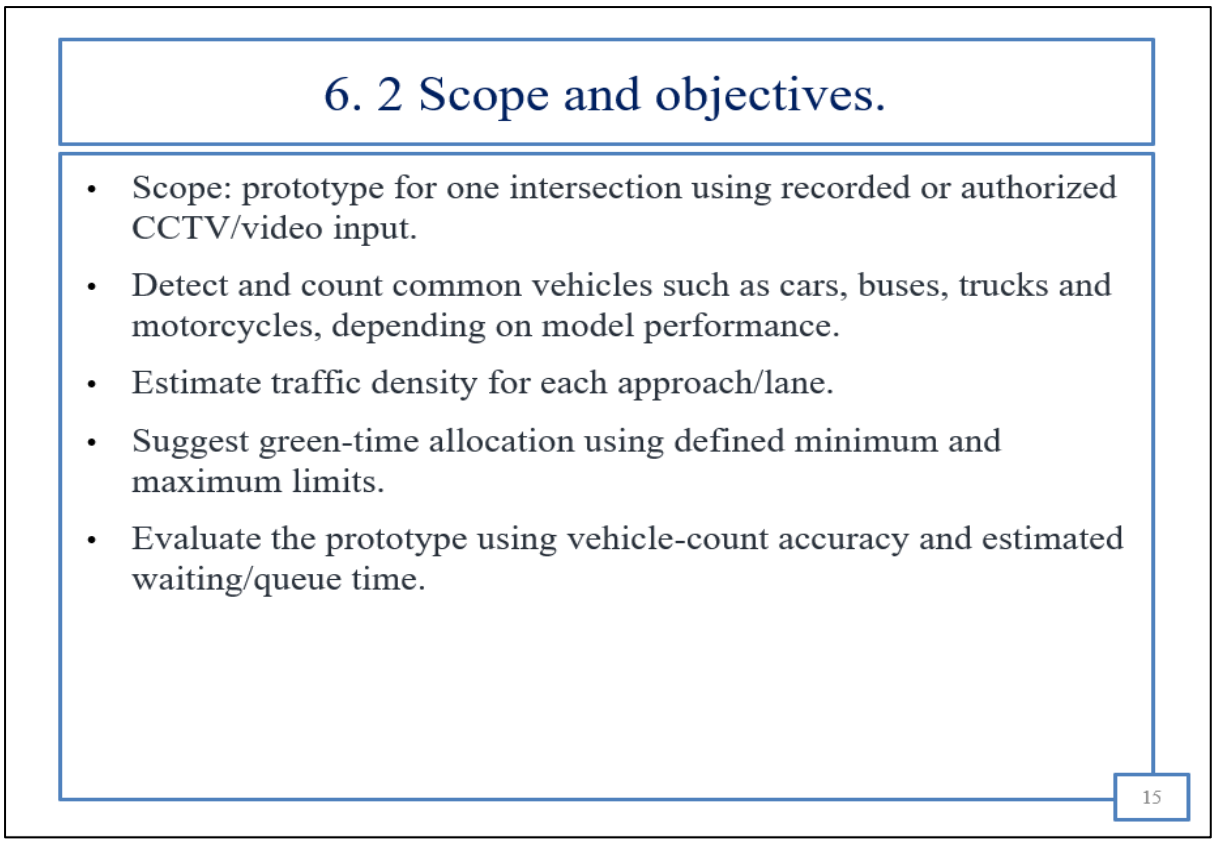


Fig No. 17: Scope and Objective

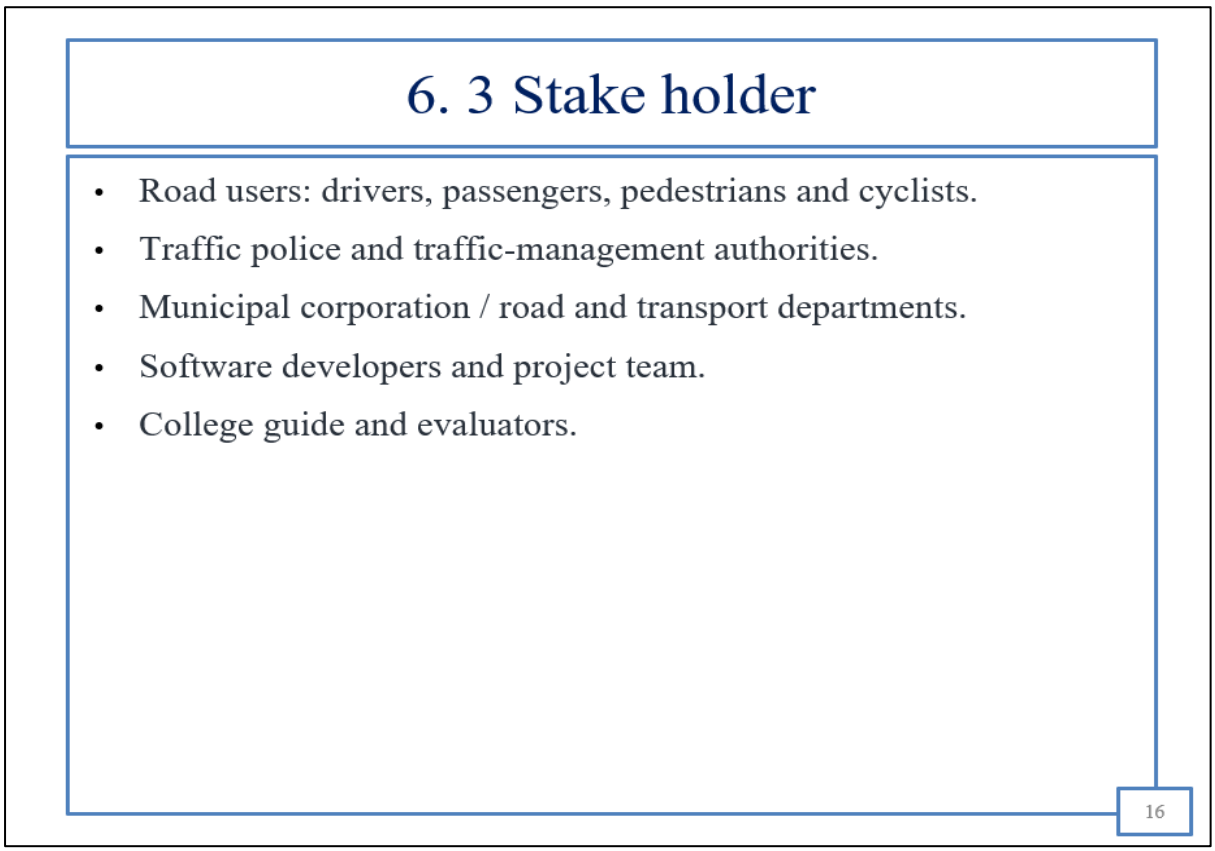


Fig No. 18: Stack Holder Page

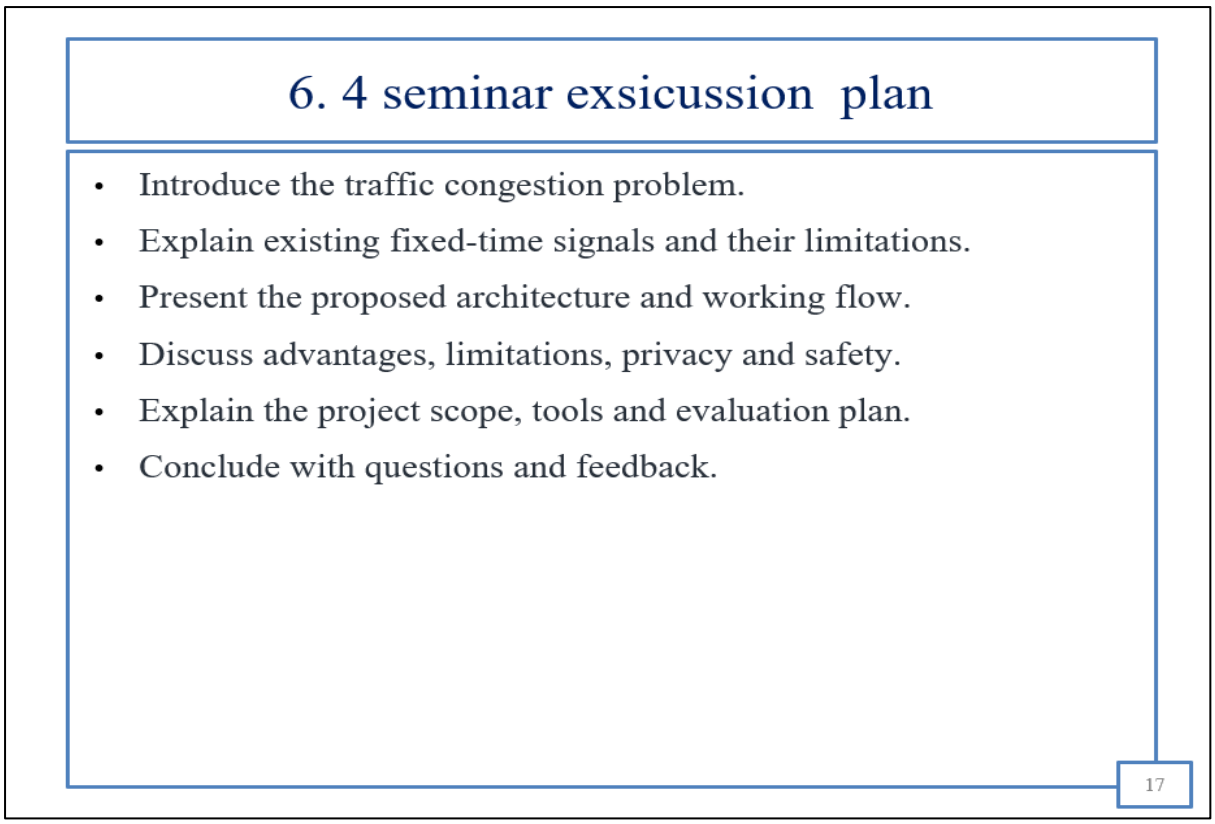


Fig No. 19: Seminar Exsicussion Plan

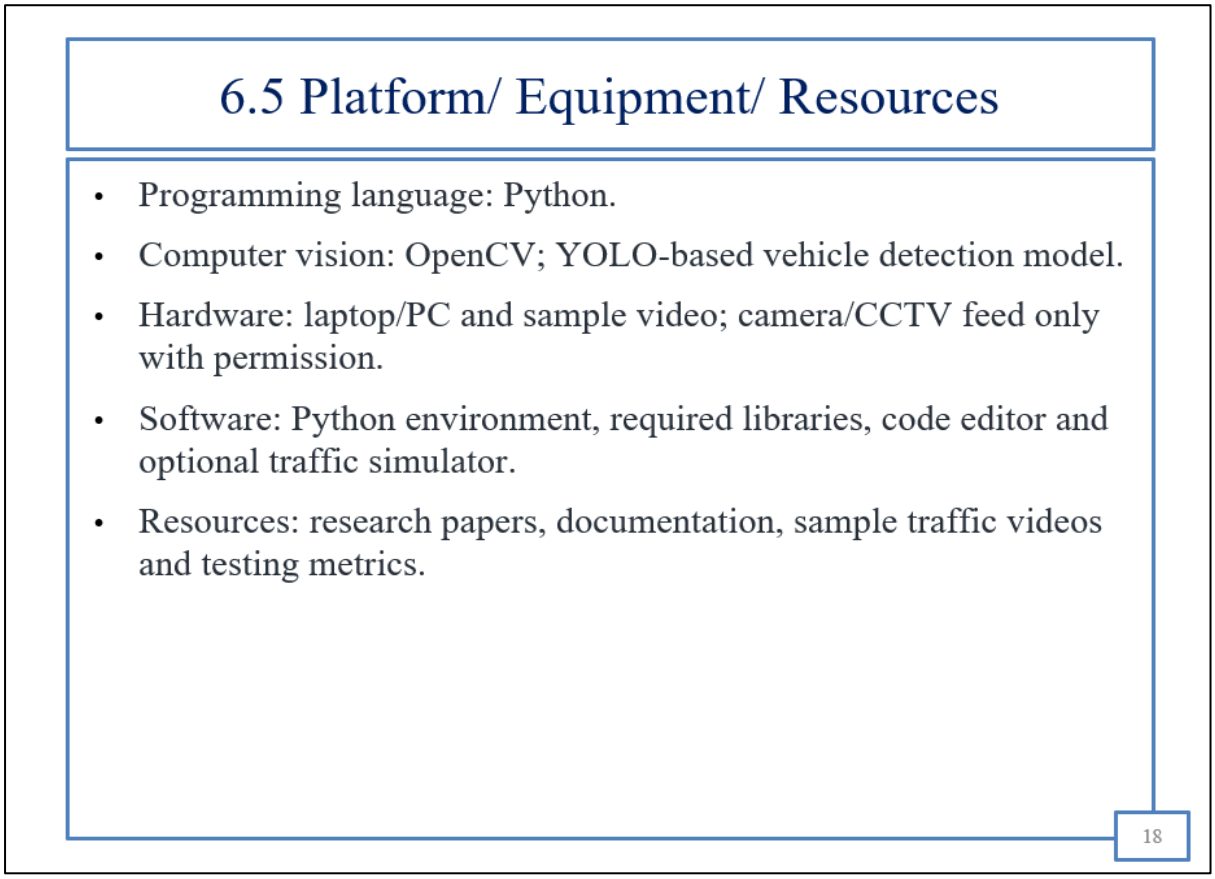


Fig No. 20: Platforms/ Equipment/ Resources

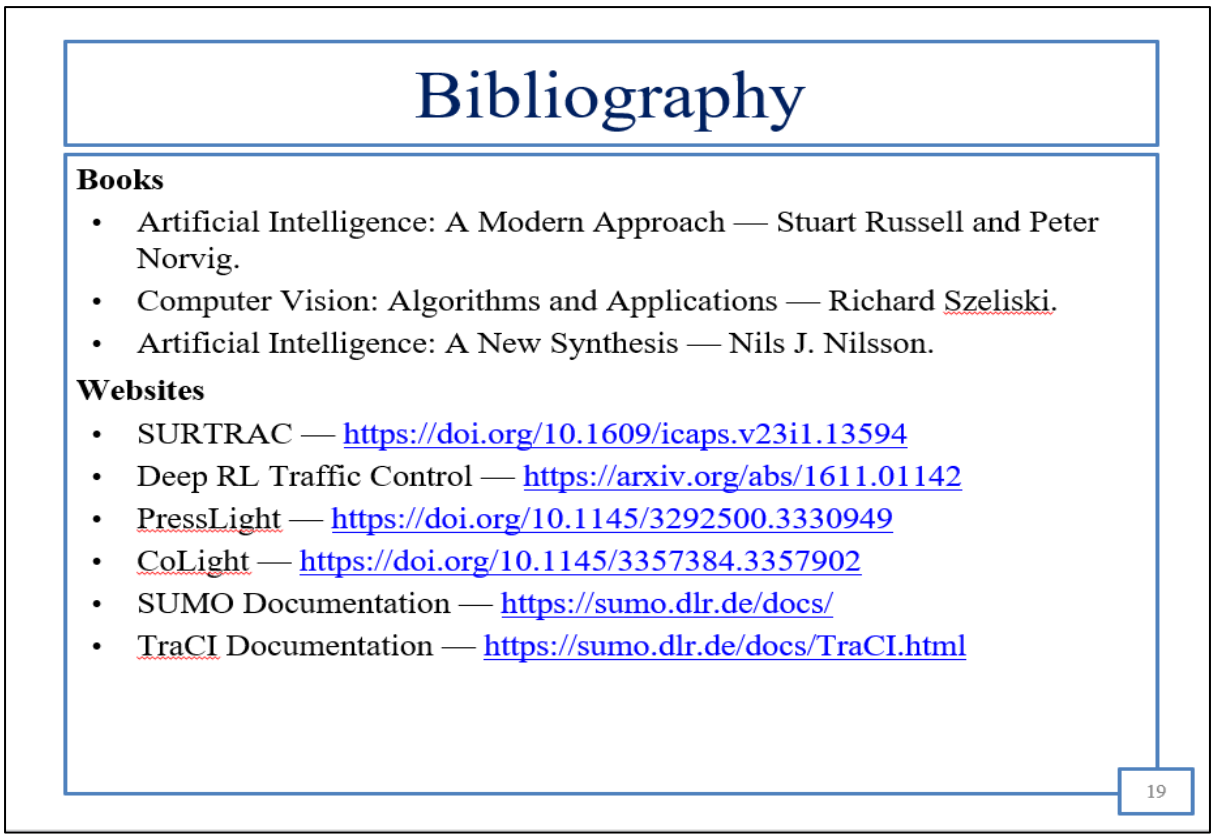


Fig No. 21: Bibliography Page

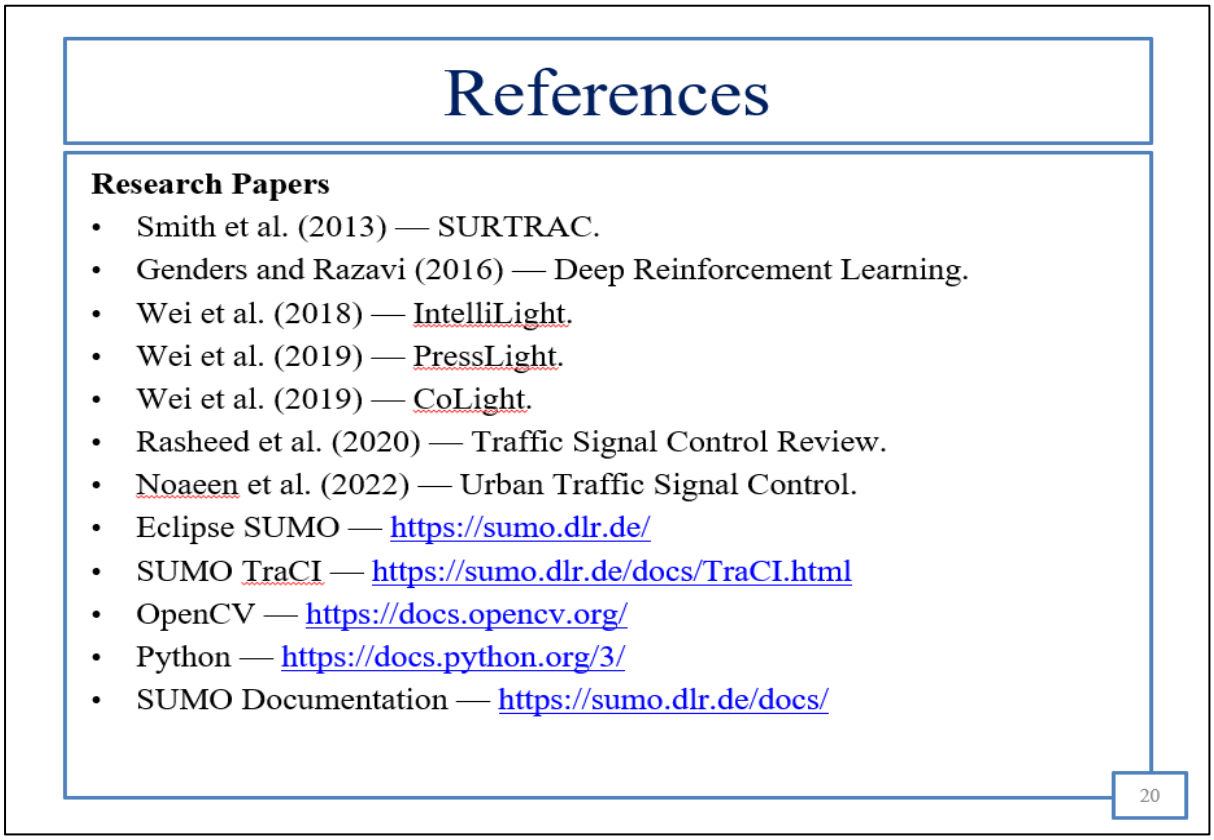


Fig No. 22: References Page

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2. H. Wei, G. Zheng, V. Gayah, and Z. Li, "PressLight: Learning Max Pressure Control to Coordinate Traffic Signals," ACM SIGKDD, 2019. https://doi.org/10.1145/3292500.3330949

3. H. Wei et al., "CoLight: Learning Network-level Cooperation for Traffic Signal Control," ACM CIKM, 2019. https://doi.org/10.1145/3357384.3357902

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5. M. Gregurić et al., "Application of Deep Reinforcement Learning in Traffic Signal Control," *Applied Sciences*, vol. 10, no. 11, 2020. https://doi.org/10.3390/app10114011

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SEMINAR and PROJECT INITIATION Assessment / Evaluation

Sr. No	Enrollment Number	Seat Number	Name of student
1	24610960251	388556	Patil Ritesh Pramod
2	24610960253	388558	Patil Sahil Yogesh
3	24610960280	388580	Sonawane Bhumika Prashant
4	24610960296	388595	Wagh Mansi kailas

Formative Assessment

Sr. No.	Criteria	Weightage	Exam Seat Number			
			388556	388558	388580	388595
1	Selection Topic / Theme of seminar	5				
2	Literature review and data presentation	5				
3	Quality of Preparation and innovativeness	5				
4	Q-A handling	5				
5	Time Management	5				
6	Seminar Presentation report	10				
7	Selection of Theme of Problem Statement and its innovativeness	5				
8	Stages of development of Action plan	5				
9	Prototyping	5				
	Total	**50**				
	Scaled to 25					

Sign :

Name : Prof. N. D. Patel

(Course Expert/s)